

Institute of Actuaries of India

Subject SA4: Pension & Other Employee Benefits

October 2015 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the paper setters with the aim of helping markers of scripts so as to have a framework and be consistent while evaluating answers. The solutions given are only indicative. It is realized that there could be other points as valid answers and the marker may give credit for any such alternative approach or interpretation which the marker considers to be appropriate.

Solution 1:**i) Different ways of considering cost of a Defined benefit scheme by a sponsor:**

Sponsor of a Defined Benefit scheme can consider cost in a number of ways as below:

- Expected long term cost of the scheme
- Cash contributions needed each year to meet this long term cost
- Impact of the pension scheme on the company accounts

The sponsor will consider each of these aspects of cost and for some employers the impact on the accounts will be critical.

[2]

ii) Accounting concepts for reporting pension costs:

The accounting concepts which are most relevant for reporting pension costs are as follows:

- Prudence:
 - Presenting an unduly optimistic set of results should be avoided
 - Prudence should only be applied in situations where there is uncertainty
- Going concern:
 - It is assumed that the business is not going to be wound up (unless it is)
- Accruals:
 - Costs are recognised when they are incurred rather than when they are actually paid
- Consistency:
 - There is comparable accounting treatment of like items from year to year
 - Or if there is not, the change is disclosed and the effect shown
- Materiality:
 - Some approximations may be made for certain costs rather than waste time calculating more precise figures
 - What is, or is not, material however does depend to some extent on the emphasis which the company (and / or shareholders) will put on the relevant figures

[5]

iii) Examples for two accounting concepts:

Accruals: a classic example for application of this concept for the purpose of accounting pension costs is that of long service awards. Typically in the Indian context, benefits under a long service award are usually paid on completion on certain service milestones during the tenure of employment. The liability for such benefit payable needs to be accounted for from day one of an employee becoming eligible allowing for the probability of him remaining in service in the future and becoming entitled to receiving the benefit applying the principle of accruals.

Materiality: For a global multinational present in multiple countries, materiality thresholds usually apply for reporting of the pension costs of their subsidiaries in different countries. The materiality thresholds may be used to determine reporting requirement (to be reported or not), methodology to be used (accrued/ accounting for example), periodicity of reporting (quarterly, annual, bi-annual etc.), due diligence to be applied etc.

[3]

iv) Report to the management of the Company:

Note: Following are the points along with the outline on the basis of which the report will be prepared

Introduction

The Company is purchasing a business unit of the Indian arm of a Global technology company and the employees are expected to be transferred to the new legal entity being set up in India. As per the terms of the sale and purchase agreement, employees would be entitled to continuity of service and hence the Gratuity benefit entitlement will transfer for the in-scope employees.

Objective

The objective of this report is to provide the acquiring company with calculations, so the Company may determine an appropriate bulk transfer value for members transferring from one scheme to another in respect of the Gratuity benefit and the long service award benefit. This would help the company also to understand the day one accounting impact, the liability numbers and its materiality, besides the estimated cost implications over the next year

The Report

- a) To achieve this objective and to provide a better understanding to the Company, I shall provide following information in the report for the Gratuity and long service award benefits:
 - i) Liability estimate for the transferring employees who are currently entitled to Gratuity benefit without ceiling and the long service benefit as per current benefit provisions as at the transfer date
 - ii) Accrued Benefit Obligation and accrued benefits as at the transfer date
 - iii) Projected expense for the next year
- b) As the continuity of service is being provided to the transferring employees hence the entire past service liability in respect of these employees shall transfer to the Company. Projected benefit obligation represents the equitable interest for these employees as on transfer date. I understand that assets for the Gratuity scheme will also get transferred and the funding ratio as at transfer date for the transferring employees shall be same as the overall funding ratio as per the latest actuarial

valuation report of the seller (any other reasonable assumption like asset equal to liability is also acceptable).

- c) The valuation as at transfer date is done on the basis of the membership information provided by the company as at a particular date. I shall value the Gratuity benefit as per the payment of Gratuity Act provisions without any cap and the Long Service Award benefit as is currently applicable to the transferring employees assuming that the same benefit will continue to be provided in the new entity. As no intimation is given for any discretionary benefits that have been provided for the transferring employees and hence shall not make any allowance for such discretionary benefits. If company is able to provide history of past discretionary benefits, then suitable allowance will be made for them.
- d) For estimating the expense for 2016, I shall assume that assets will also be transferred against the liability transfer for these employees.
- e) I understand that the seller will transfer assets to the buyer on the basis of the overall funded ratio of the existing gratuity fund. Interest will then be added to the above figure at 8.5% p.a. (an example only) for the number of months taken for the actual transfer of assets. Any interim benefit payments need to be paid directly from the company and adjusted in future contributions payable to the trust. However, for reporting purpose I shall consider the asset value as if the amount would be received as at the transfer date (any other reasonable assumption is also acceptable). I shall not make any allowance for any likely difference in assets transferrable due to movement in fair value of the assets.
- f) For projecting the expense for 2016, I shall consider the expected contribution to be equal to the expected benefit payments for the calculation of Expected Return on Assets (any other reasonable assumption is also acceptable).
- g) Following declarations/definitions shall form part of the report:
 - i) I have not considered the impact of any other legislative changes, judicial pronouncements, practices and tax regulations in the future for the purposes of this report.
 - ii) The membership data considered is as furnished by the company and all the assumptions used for valuation purposes along with the plan provisions have been approved by the Company. In preparing this report I have relied on the completeness and accuracy of all the data and information provided to me orally and in writing by or on behalf of the Company and its advisers. I have not completed any detailed validation checks on the information provided. I have,

however, carried out broad consistency checks. The data, assumptions and plan provisions used are set out in subsequent sections of the report.

- iii) The assumptions used for the purpose of valuation are as set out in the Actuary's letter in the sale and purchase agreement. These assumptions are reasonable for the purpose of this valuation.
- iv) I have used the Projected Unit Credit (PUC) actuarial method to assess the Plan's liabilities, including those related to death-in-service and incapacity benefits.
- v) Under the PUC method, the Actuarial Liability for active members either as at the valuation date is calculated taking into account all types of decrement. In such calculations pensionable pay is projected from the relevant date up to the assumed date of retirement, date of leaving service or date of death as appropriate.
- vi) All figures in this report are in local currency i.e. Indian rupees

The results for Gratuity Benefit Schemes as at the transfer date are given below:

Approach	Actuarial Liability		
	Gratuity Benefit	Long service benefit	Total
Projected Benefit Obligation (PBO) as at transfer date			
Accrued Benefit Obligation (ABO) as at transfer date			
Accrued Benefit as at transfer date			

**The accrued benefit as at transfer date includes the accrued benefit for unvested employees.*

***The assumptions used to arrive at the liability as at transfer date have been summarized in the Actuary's letter that forms part of the sale and purchase agreement*

**** the basis used for the purpose of US GAAP and Indian GAAP are identical and hence the PBO is the same under both GAAPs*

The asset value as at transfer date to be transferred from the seller to buyer is INR XXX.

The projected expense for the year ending after close date under different standards is shown below:

Approach	Gratuity/ Long service award		
	AS15(rev)	IAS19(rev)	USGAAP**
Current Service Cost	X	X	X
Interest Cost	Y	Y	Y
Expected Return on Plan Assets	(Z)	(Z1)	(Z)
Curtailment Cost/(Credit)	0	0	0
Settlement Cost/(Credit)	0	0	0
Past Service Cost	0	0	0
Actuarial Losses/(Gains)	0	0	0
Total Employer Expense recognized in P&L*	XX	XX	XX

**The projected expenses for the year ending after the close date stated above includes only indicative numbers for interest cost and service cost. The actual expense for this period would change depending upon the projected benefit obligations as at the end of the year, actual benefit payments and the actuarial gains/losses that would arise in the year ending after the close date..*

*** I have assumed that under US GAAP no transfer of Accumulated other comprehensive income in respect of these employees will transfer to the buyer.*

1. Projected benefit obligation is the actuarial liability in respect of the past service rendered by each member at the date of valuation.
2. The expected return on plan assets over the accounting period is based on an assumed rate of return as set out in the Actuary's letter. Under IAS19 (rev) norms, the assumption of expected return on assets is equal to the discount rate.

3. I have assumed that salary increases take place at the middle of the year for projection of benefit to normal retirement age/exit
4. I have assumed that all decrements will take place at middle of the year and retirement will take place at the beginning of the year.

The key differences between AS15 (revised) and IAS19 (revised) 2011 may be summarized as below:

- Under AS15 (revised), gains/losses are immediately recognised directly through P/L statement. Whereas under IAS19(revised) 2011, gains/losses are recognized through OCI (Other Comprehensive Income)
- Like-wise when past service cost/ credit (unvested portion) is amortised over the remaining vesting period under AS15(revised), the impact of any such changes (total past service cost, curtailment and effect of non-routine settlements) have to be recognized immediately in the P&L under IAS19(revised)
- As mentioned above under IAS19, the rate of return on assets is based on the discount rate instead of a separate expected rate of return on asset assumption under AS15(revised)
- Also IAS19 (revised) requires enhanced disclosures that are focused on risk and is principle based. For example, Additional disclosures as needed to meet the disclosure objectives e.g. nature, characteristics and risks associated with the PBO, e.g. split between categories of membership

[30]

v) Factors contributing to the liability accrual:

The Gratuity benefit as per current statutory provisions has a cap of 10 lacs overall on the benefit payable. However as per the benefit terms, no cap is applicable under the proposed Company scheme. Being a final salary linked defined benefit liability, having no cap would introduce a significant amount of volatility and unpredictability in the liability accrual going forward. The experience of actual salary increase vs. assumption used for estimating the liability would be a key contributor to the liability gain/ loss going forward. For instance if the assumption used is 8% p.a. and the actual salary escalation is 10% p.a., the difference of 2% would result in a liability loss.

Likewise difference between assumed and actual attrition would impact the liability accrual as well. For example, if unvested attrition assumed is 10% (i.e. attrition of employees before completing the vesting criteria of 4 years and 240 days) and the actual attrition is only 6%, then there would be an experience loss.

Also change in the assumption basis going forward would impact the liability accrual too. A change in discount rate basis would either increase or decrease the liability estimate as at a particular date depending on whether the discount rate has reduced or increased over the accounting period respectively.

[3]

vi) Accounting implication of capping future accruals:

If the Company decides to cap the Gratuity benefit going forward, the accounting liability would reduce as the benefit terms would be comparatively lower.

The reduction in accounting liability as on effective date of change would constitute the past service credit. For example, if the change is done, say with effect from 1 January 2016 and the accounting liabilities using the same set of assumptions based on payment of Gratuity Act provisions without and with a cap work out to, say INR X and INR Y, then X-Y would be positive and will be a past service credit as on that date.

The unvested portion of the past service credit needs to be amortised over the balance vesting period under AS15 (revised). However under IAS19 (revised), the entire past service credit will flow through P&L.

In terms of estimating the past service credit, care must be taken to ensure that the accrued benefit is not reduced even if in some cases the same has exceeded the 10 lac cap applicable under the statutory provisions. In such cases, allowance must be made only for no further accruals in the future for the purpose of estimating the credit due to the introduction of the cap.

[7]

[50 Marks]

Solution 2:**i) Commutation of Pension [Section 10(10A)]:**

- In case of employees of Central & State Govt., Local Authority, Defence Services and corporations established under Central or State Acts, the entire commuted value of pension is exempt.
- In case of any other employee
 - If the employee receives gratuity, one third of full value of commuted pension will be exempt from tax under section 10(10A) (ii) (a).
 - If the employee does not receive gratuity, one half of full value of commuted pension will be exempt from tax under section 10(10A)(ii)(b).

[2]

ii) Factors to be considered while setting out commutation factors:

1. Should be fair & equitable to all classes of members of the scheme.
 - it means

- a. For members who exercise the option, Commutation factor should represent the fair value of the pension commuted in units – (example Re.1/- per month)
 - b. Members who are not exercising the option should not be disadvantaged by the decision
 - c. Should be fair to employer/Trustees – it means factors should not cause any stress to the fund or any additional cost to the employer
2. Market consistent approach – will be fair & cost neutral- discount rates could be based on market yields available on Government long term bonds– but there will be difficulty in placing the value for the longevity risk – factors will be volatile & members retiring at different times will be getting different commuted value – Trustees will also need to arrive at the factor each time a member exercises the option
3. A long term approach is the practical way of developing the factors and this approach will be fair & consistent over a period of time:
 - a. Discount rates should be based on the likely investment return to be earned by the pension fund over a longer period with due allowance to be given for the possible variations – allowance should be given for possible rise/fall in the investment return –should be based on the investment strategy adopted by the Trustees & investment guidelines for pension funds
 - b. To be fair to members, longevity used in commutation factors should reflect the expected longevity of the pensioners of the scheme as a whole. A suitable Pensioners' mortality table should be used after due modification, if any to reflect this
 - c. The proportion of members likely exercising the option may be based on industry data
4. Presentation of factors:
 - a. Age related factors may be used– but members retiring at different ages will be getting different commuted values for the same amount of pension commuted

- b. A single factor for all ages can be easily understood by members, particularly workers but this factors has to be reviewed more frequently to ensure its cost neutrality
 - c. Can consider developing factors for age bands
 - d. Different factors for female/male members may be more fair to members and the employer/Trustees but can be viewed as discriminatory by members or by local legislation
- 5. Should the factor include family pension? This will be dependent on the Scheme Rules
- 6. Need to decide on the frequency for revision of factors. Trustees may retain the right to revise the factors after giving due notice to ensure reasonableness, practicality, fairness and cost neutrality
- 7. Other factors to be considered:
 - a. Special rates for impaired pensioners?
 - b. Other Scheme Rules – e.g. Restoration of full pension after some years
 - c. Other lump sum retirement benefits received by the employees
 - d. Limit imposed by Regulations, say, Income Tax Act/ Rules
 - e. Factors used by other companies
 - f. Increases in pension, if any

[10]

iii)(a). Professional guidelines to be followed by a Consultant Actuary in case of appointment of a new Advisor:

1. Professional Conduct Standards (PCS) sets the standards to be followed when taking new assignments and on commenting the role of scheme actuary
2. The Company/Trustees have the indisputable right to choose and to change professional advisors, or to take a second opinion and/or to have separate advisors for different matters
3. The consultant *actuary who is* invited to give *advice* to a client for the first time should contact the earlier actuary, *if any advising the client and inform him of the assignment he proposes to undertake seeking his opinion whether there are any*

professional reasons not to accept the appointment or any particular considerations which ought to be borne in mind before giving advice

4. *The earlier actuary*, upon receiving such an enquiry, must provide the required information as soon as possible and should make no attempt to obstruct the client who wishes to appoint advisors
5. *Both the actuaries* must avoid any action that would unfairly injure the professional reputation
6. Criticism of one member's work by another *member* is acceptable, provided that the criticism is properly reasoned and believed to be justified
7. Where criticism of an actuary's work is made, care must be taken to acknowledge that the other actuaries may quite properly hold different professional opinions and that special circumstances may exist in any particular case
8. There is no conflict of interest in taking up the assignment
9. The actuary holds the certificate of practice, if any required for the assignment
10. The actuary has necessary expertise and experience for the assignment

[4]

iii)(b). Information to be called from the Employer/Trustees:

General: Information to be called should be complete & sufficient to cause an investigation into the scheme within the time frame

Scheme data: This includes Scheme Rules, Information provided/Notices issued to members including Scheme Booklet, Benefit Illustrations, if any, Enrolment Forms etc. This will provide information about the scheme benefits payable for different categories of members, eligibility criteria, normal retirement age/s, options to retire early/late, financing, contribution requirements, process for changing the rules etc.

Trust Deed & Deed of Variation, if any: It will specify the roles & responsibilities of Trustees. Powers/Duties of Trustees (For example, can the Trustees alter the benefits or provision for any discretionary benefits). Approval obtained from Commissioner of Income Tax

Scheme Members' data: To determine the liability of the scheme as given in the following table:

Active members	Pensioners	Spouse/ Family pensioners
Name, Gender, Marital Status	Name, Gender, Marital Status	Name, Gender, Marital Status
Identity number	Identity number	Identity number
Date of entry into service	Date of Birth	Identity number of deceased/member pensioner
Date of entry into Scheme	Date & cause of Exit	Date of birth/age of family pensioner
Date of birth	Date of commencement of Pension	Relationship with member
Pay	Eligible Pension	Date of commencement of family pension
Pensionable Pay (if different)	Commuted Pension	Family pension received
Contribution to pension scheme	Pension received by member	

Asset Data: Details of assets held in the fund such as Government Bonds, Corporate Bonds, Equities with their face value, market value, coupon rate, original term, maturity date etc. If it is not possible to provide the complete list of instruments held individually, the asset data classified into the broad categories with outstanding terms. The credit rating of the bonds, charges/encumbrance on the assets, details of self-investments, if any. Details of short term deposits such as Bank deposits, Mutual Fund investments etc. should also be provided. If the scheme has any investments with life insurers, then their details, such as type of plan (whether Unit-linked or Deposit Administration) details and their value at end of period. A copy of policy document should also be called for. The surrender or realisable value should also be collected

Actuarial reports: For last financial year which will include the details of the members' data used, method & basis used to value the assets & liabilities, results of the actuarial

investigations, analysis of surplus, other experience analysis, if any done by the earlier actuary etc.

Financial statements: of the company and the scheme for the last year – this will give overall financial strength of the company, significance of the pension cost to its overall expenses, pension cost measured & recognised as per AS 15 (Revised), any unfunded pension liabilities charged to balance sheet, if any.

Data relating to the future operation of the scheme: This will include:

- The company's future business strategy for expansion, diversification, mergers & acquisitions etc. This may indicate how the financial health of the company is likely to vary over a period of years
- Likely changes in the salary levels and/or other employee benefits being offered. Any changes in the benefit structure likely to have material impact on the pension scheme– for example, changes in the Pay structure may require alteration in the Pensionable Pay; changes in benefit structure without increasing the employer cost may require restructuring of pension & other benefits etc.
- Changes in the recruitment/retrenchment policy in the coming years- whether any redundancy exercise is being considered or mass recruitment programme is under consideration
- Changes in Pension Scheme Rules under consideration– the employer may have some objective of conducting this exercise; A discussion with the company's management may help to understand their concern/objective
- Any other information that may be relevant for this exercise.

[10]

iv)(a). Reasons for increasing the commutation factors:

Company may be concerned with the increasing cost of providing pensions.

The increasing cost may be mainly attributed to:

- a. The decreasing investment returns of pension fund (8.75% to 8.70% over the period of last 5 years)
- b. Increasing longevity in general experienced by many pension funds. (Though no scheme specific experience available)

The average age of active members increasing fast and their number has grown by 2.5% in five year period. The number of pensioners, though small but growing fast (grown 200% in last 5 years). The proportion of members approaching retirement (age group 50 -60) has increased from 5% to 11% and the proportion of middle aged members (age group 30 – 50) increased significantly from 44% to 55%

Though no information is available in respect of the proportion of members commuting less than the maximum permitted pension, the Company may want to de-risk the pension fund. It may like to encourage members to commute maximum permitted pension to avoid the impact of increasing cost of providing pension from the fund

By increasing the commutation factors but still keeping the factors sufficiently below the cost of providing pension, the company may aim to strengthen the fund

There may be some miscellaneous reasons such as:

- Availability of surplus in the fund
- Availability of liquid assets
- Generosity of the employer to pay increased benefit
- To match factors offered by other schemes

[4]

iv)(b). Views of members of different categories on the proposal and its impact on the funding level of the scheme:

This is a contributory pension scheme to which members contribute 3% of their pensionable pay per annum. Members leaving before age 50 will continue to view the scheme as unreasonable/unfavourable. The contribution of this group of members has been an important source of surplus. They are not getting any benefit out of the scheme. These members will expect revision in scheme rules.

A close look at the old and proposed commutation factors shows that the factors have been increased by 10 % at all ages.

Assuming the revised factors are still below the cost of providing pension, the exercise appears to aim to continue to generate surplus, though at the same time encouraging commutation. This may ultimately improve the security of the benefits of all categories of members in general.

But existing pensioners may expect share in the surplus as they have got lower commuted value and hence have contributed significantly to the surplus. They may expect either one time or periodic pension increases.

Assuming spouse pension is not included in the commutation, family pensioners may not be affected by the proposal.

Active members who are approaching the retirement will view the revision of factors favourably, though may expect further increase if factors under other schemes are higher.

Generally members commute to the extent maximum permitted at retirement due to various reasons such as to meet the lump sum obligations, e.g. repayment of housing loan and/or to avail the tax free lump sum. Due to these reasons most of the members will continue to commute to the maximum.

The proportionate number of retiring members opting commutation of pension at different ages is likely to increase in future. This may cause some cash flow problems for the Trustees.

[5]

iv) (c) Other ways of utilising surplus:

1. Improvement in benefits:

- a. To leaving members before age 50 - for e.g. return of contributions with or without interest which may be viewed removing anomaly and fair treatment to all classes of members
- b. Increases in benefits to pensioners & family pensioners
- c. Relaxing the eligibility criteria, e.g. allowing pension benefit on retirement before age 50
- d. Increase the target pension, e.g. increase the accrual rate
- e. Introduce minimum pension
- f. Introduce child pension up to certain age, if not already
- g. Introduce generous impaired life pension(or lump sum) to retiring members subject to medical examination
- h. Introduce/enhance death cover for active members – taking a GTL cover is an attractive way of using surplus which active members are likely to appreciate

2. Reduction in Contribution
 - a. Contribution holiday for the employer
 - b. Take refund from fund (but it has tax implications even if permitted)
 - c. Reduction of contribution from members
3. Changes in the investment strategy, e.g. increase proportion of equity investment to the extent permitted. It may help in increasing surplus over the long run
4. Charge expenses to the fund, e.g. professional fees, salary of Trustees
5. Leave the surplus in the fund as the surplus is small which may take care of future deficits, if any

[5]

v) (a) Points to be covered in the reply to the pensioner:

As the scheme is a defined benefit scheme, the members are entitled to receive their eligible pension from the scheme as per rules of the scheme.

The members are entitled to commute a proportion of their pension only at the commutation factors applicable at the time of commutation.

As the member has already commuted to the maximum possible to extent, further commutation cannot be allowed as it will be against the Rules of the scheme which are approved under Income Tax provisions

It is not appropriate to compare the commutation factors offered by the scheme with the premium rates at which the annuity product is being offered by the insurers. Had it been a defined contribution scheme, then the retiring members would have got the right to compare annuity rates offered by insurers and the annuity rates at which Trustees are converting retirement pot of members into pension.

Since the pension product launched by the Government through life insurers might be at subsidised rates, it is not fair to compare the premium rates of such pension product with the commutation factors offered by the scheme.

Commuting pension after commencement of its payment to the member is considered as surrender which is not permitted as a member whose health deteriorates would always want to surrender his pension for a lump sum.

Therefore the request of the member is to be declined.

[5]

v) (b) Additional points to be considered in the case of ailing pensioner:

Most of the arguments as stated in 5(a) relevant in this case also. The Trustee need to consider the special circumstances of the individual pensioner while making request for commutation after he has retired.

The pensioner is entitled to receive pension during his life time which is expected to be much shorter than that assumed by the pension scheme on an average. However, there will be few others who will receive their pension for a much longer period than that assumed.

Whether or not he may be allowed to commute his future pension- either full or part will depend upon the following:

- Type of pension being paid. If it is payable for life with return of capital on death of the pensioner, then commutation may be allowed
- If it is payable for life only, then commutation cannot be allowed
- If it is payable for a period certain (say, 15 years) and thereafter for life, then commutation of pension for unexpired duration can be permitted
- Trust deed & Rules of the scheme need to be referred to ensure whether commutation in such extra ordinary circumstances may or may not be permitted. It may need permission of CIT
- Such commutation, even if allowed will be subject to deduction of tax

[5]**[50 Marks]**
